Motors and Actuators

Actuators

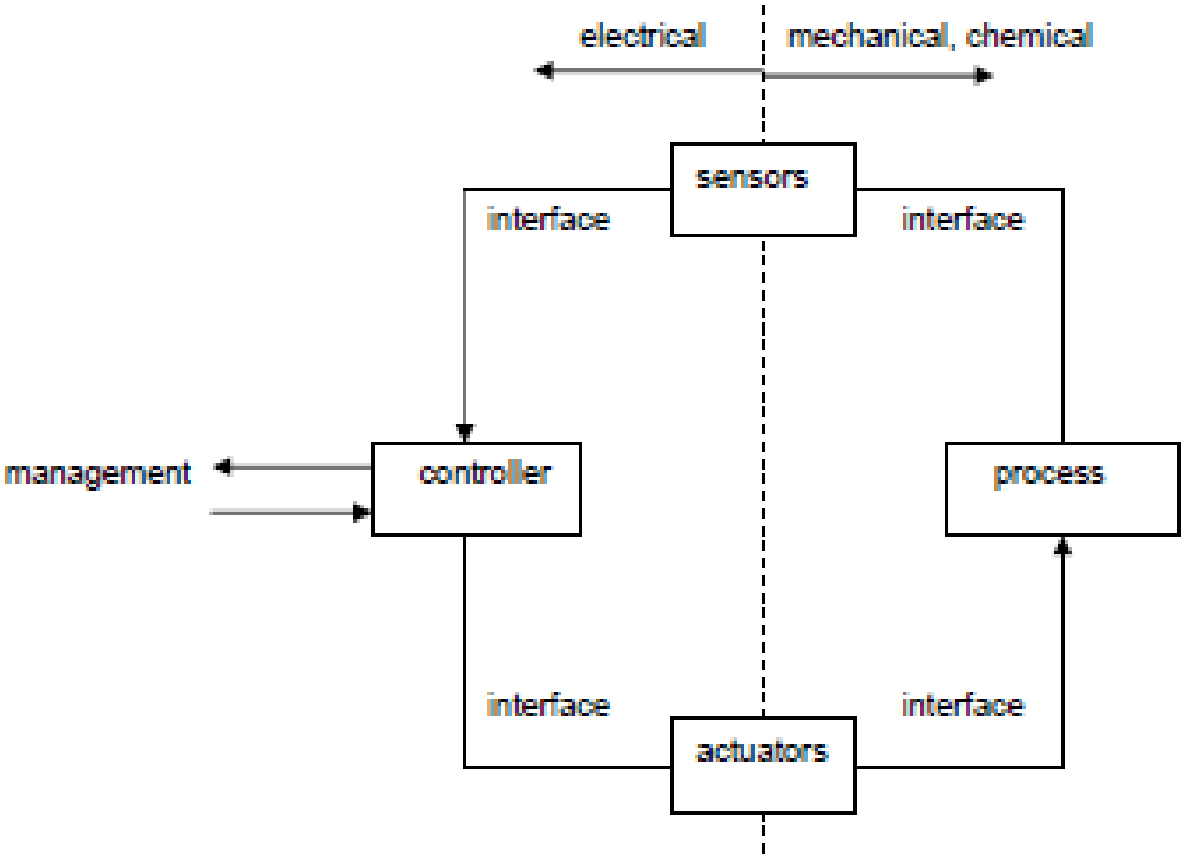


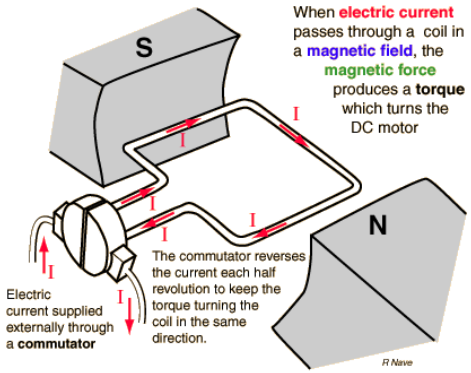
Fig 1: Interconnections in a typical mechatronic system

Actuators

- Actuators are a too broad a category to cover completely in this course, but small electric motors are a common tool used in the Electronics and Mechatronic design engineers “toolbox”. We will cover four such motors in more detail:
  - Commutated DC motors
  - Brushless DC motors
  - AC induction motors
  - Stepper motors
- We’ll also make small mentions of linear actuators and switches as well as micro mechanisms.

Commutated DC Motors

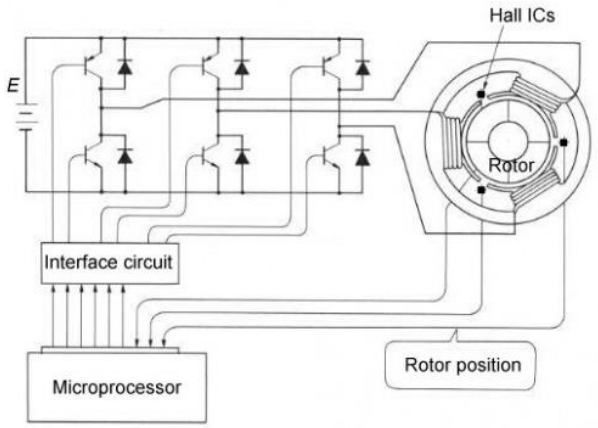
- Very common small motors used in many everyday applications (shavers, cordless screwdrivers, etc.)
- In most cases the field is provided by a permanent magnet, and thus their direction can be reversed easily by reversing the voltage
- These motors are often used in some sort of position feedback control using a pot. In this format they are commonly called servo motors



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Brushless DC Motors

- Commutator has been replace by an electronic system which senses the shaft angle and switches the field current accordingly
- For high reliability and so are used in systems which must run for very long lifetimes (e.g. computer fans)
- In order to avoid having electrical connections to the rotor, the rotor is generally a permanent magnet structure, which limits the torque and increases the moment of inertia



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Brushless DC Motors

- *Speed Control*
  - *These motors require a fairly sophisticated electronic system to run at* all, and it is not generally difficult to adapt this for speed control, provided one can get access to the basic signals. They can be very accurately speed controlled in open loop, but may stall if overloaded.
- *Power and Torque*
  - *These motors are not particularly good from a torque point of* view, but are reasonably powerful. They are generally considered to offer the highest efficiency when properly used (hence their widespread use in electric cars).
- *Reliability*
  - *These motors are highly reliable mechanically. The electrical system* requires protection in the standard ways.
- *Other*
  - *Currently these motors are mostly used where long life and accurate speed* control is essential (e.g. to spin disk drives) but we can expect them to be more widely used.



http://www.tedmotors.com/56.htm

AC Motors

- These motors are seldom used.
- They have two very significant features
  - an AC synchronous motor can be driven from mains and will turn at a speed linked to mains frequency (so a mechanical clock, for example, can be driven with an AC motor).
  - A rarely used advantage is that a wound-rotor AC motor can be built with very little steel (or in fact no ferromagnetic material at all), allowing their use in areas of strong magnetic fields (inside instruments such as NMR machines and electron microscopes for example).



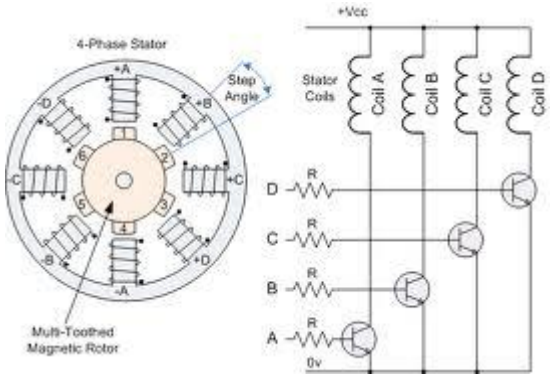
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Ac motors

- ***Speed Control:*** *Synchronous motors turn at some subharmonic of the AC* synchronous speed. They tend to stall if overloaded. Induction motors slip rather than stall, and can be voltage or frequency controlled, but may be inaccurate in open loop.
- ***Power and Torque:*** *These differ from one type to another. Synchronous motors offer* moderate power and torque, while induction motors generally have good torque for their size and weight.
- ***Reliability:*** *Induction motors can be constructed with no electrical connections to the* rotor and are therefore mechanically very reliable; motors using slip rings (which includes synchronous motors) are less reliable but are still substantially better than commutator motors.
- ***Other:*** *These motors are unlikely to be used in small-size applications, unless AC* frequency control is useful.

Stepper Motors

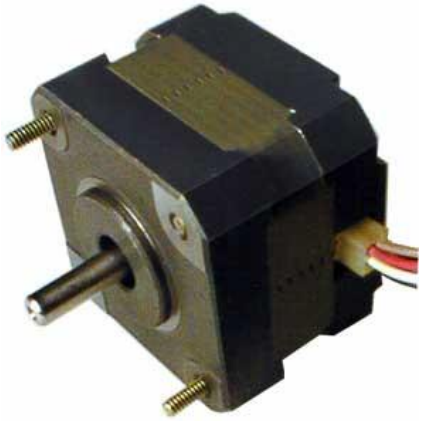
- Stepper motors are commonly used where uncomplicated position control is needed. They are commonly found in within computer cd/dvd rom drives to control the motion of the disc tray.
- Many chips exist that remove the control aspect of the many windings and convert the required user input to a simple direction input and step input.
- Have a look at the SAA1027 by philips.



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Stepper motors

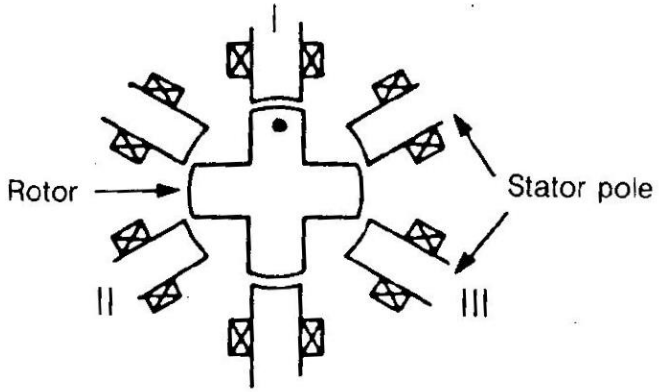
- ***Speed Control:*** *Stepper motors, like permanent magnet motors, require electronic* control for their basic operation. They can be very precisely controlled in open loop, and can be controlled at very low speeds. They stall when overloaded.
- ***Power and Torque:*** *They have very poor power and torque performance.*
- ***Reliability:*** *Stepper motors are very reliable owing to their brushless design.*
- ***Other:*** *Stepper motors are only useful when digital control is being implemented, but* they are extremely versatile in this case. They are the only type of motor which has a reasonable holding torque (resistance to being turned when standing still). They are also the only type of motor in which the shaft position is directly controllable in open loop.



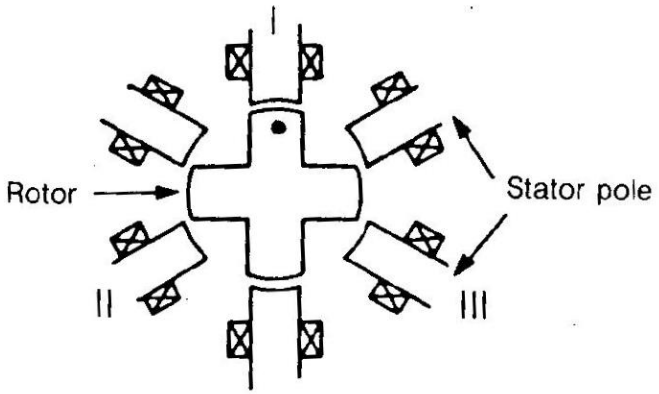
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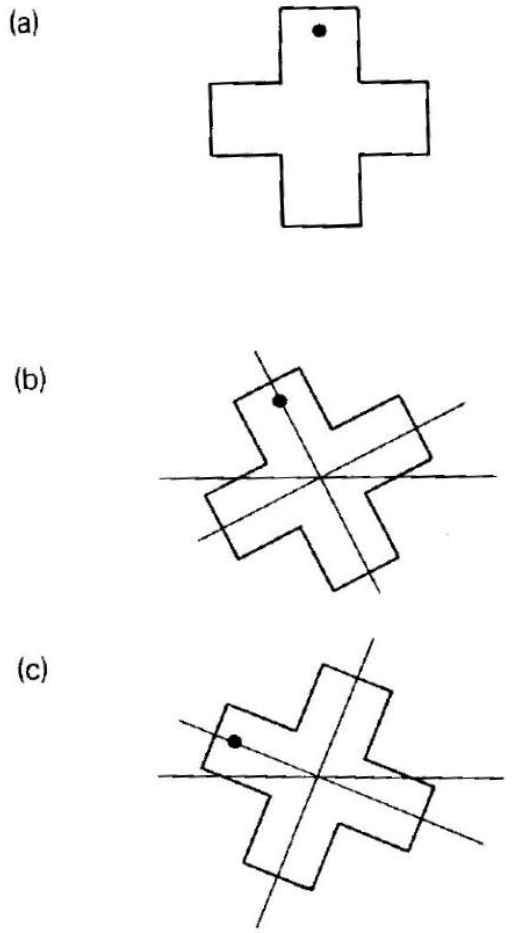
A Brief Discussion on Stepper Motors

- Stepper motors are very widely used in PC-controlled systems and are of particular interest to mechatronic designers for that reason. They generally come in three flavours: variable reluctance, permanent magnet, and hybrid stepper motors. The different types are distinguished by the material used in their rotors. The basic principle of operation (shown below for a variable reluctance motor) is the same for all types:



A Brief Discussion on Stepper Motors





Solenoids

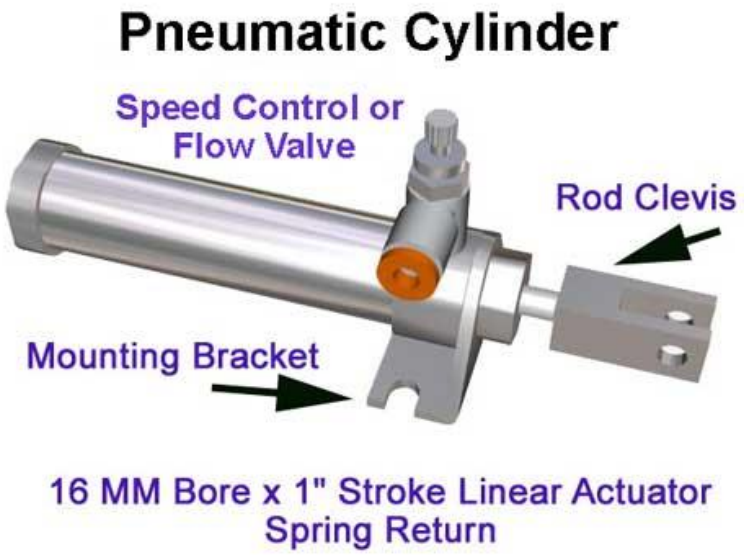
- Linear actuator
- Simple on off control
- Come in many different sizes and powers and can exert a great deal of force, but have practical limits in terms of power
- Not much speed control



http://www.fwmurphy.co.uk/images/RP23xx_solenoid_medres.jpg

Pneumatics

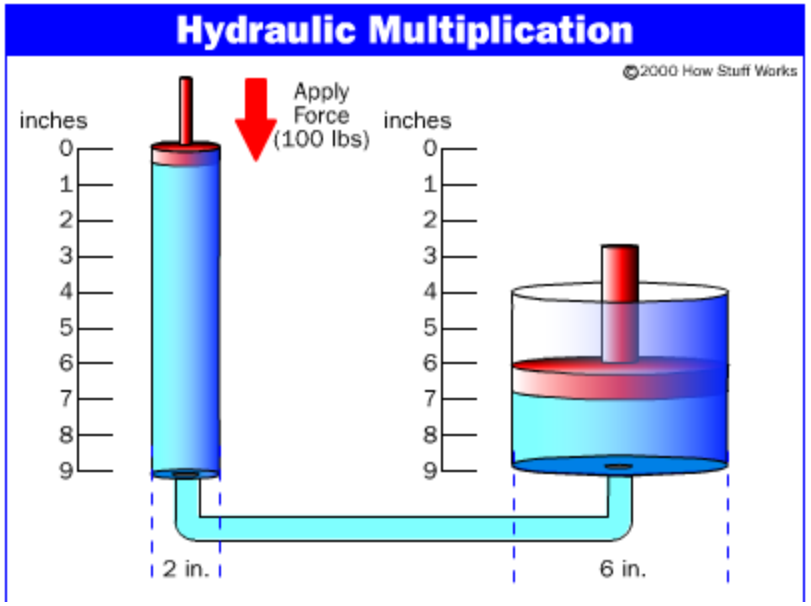
- Pneumatic cylinders provide a variety of linear actuator solutions.
- Using valves and position switches, the rate and direction of airflow can be controlled.
- Repetitive mechanical tasks can be assembled by a carefully designed system of pneumatic cylinders providing linear actuation and considerable force appropriate to each stage of the task.
- This is used extensively in industry in assembly and manufacturing
- Most Pneumatic systems use a “power supply” like that of a circuit which is a pumped up air compressor cylinder
- The cylinder provides pneumatic “potential difference and a variety of valves which resemble logic gates control the behaviour of the actuators.
- The setups are usually very expensive and limited to big industry.
- Speed and position control are limited and are difficult to implement as part of a control loop



http://www.gearseds.com/curriculum/images/figures/web_pneumatic_cylinder.jpg

Hydraulics

- Hydraulic actuators are the big cousin of pneumatics and use the incompressibility of water or normally oil to exert massive amounts of force.
- They are used in large machinery normally to control multi jointed arms to perform complex tasks requiring careful control and massive force
- Don’t forget about conservation of energy and the more force you need, the slower it must go



http://science.howstuffworks.com/transport/engines-equipment/hydraulic1.htm